Work package 3: Scoping, assessing and conceptualizing the needs and requirements of integrating AI-driven solution in the operational business of MFIN

The implementation of work package 1 has enabled the recording of customer queries during non-working hours, with call flows deployed across Hindi-speaking regions. MFIN has begun receiving these audio recordings and requires an AI/ML-based solution to classify the complaints using keywords.

The proposed solution consists of two main components:

1. **Automatic Speech Recognition (ASR)**: A contextual ASR model is identified to transcribe audio files related to microfinance, converting speech to text.
2. **AI/ML-based Classification**: Using the generated transcriptions, a machine learning model classifies the audio data into relevant categories based on specific keywords.

In collaboration with the MFIN team, we developed a solution based on technical feasibility, usability, and its seamless integration with the MFIN CRM. The report includes:

- Exploratory analysis and evaluation of ASR transcriptions to identify the best model for the specific use case.
- Experiments with different classification models to select the most suitable one for categorizing the audio queries.

Additionally, model comparison, infrastructure requirements, and integration recommendations with MFIN's existing IT framework have been detailed for optimal implementation.

1. Methodology

A. **Dataset Creation**: A dataset of audio recordings was curated by MFIN team from the queries received during non-working hours across Hindi-speaking regions in MFIN's call flows. These recordings primarily consist of complaints and queries related to microfinance, providing a diverse and representative sample for classification tasks.

B. **Audio Transcription Using ASR**: The audio recordings in the dataset were transcribed manually by moderators and also using Automatic Speech Recognition (ASR) models. Multiple ASR models were considered, and each transcription was generated using speech-to-text services tailored to the context of microfinance.

C. **Exploratory Analysis and Ground-Truthing**: An exploratory analysis was conducted on the transcriptions to assess their accuracy and relevance. Ground-truthing was performed to ensure the transcriptions reflected the actual content of the audio files. This step helped in selecting the best ASR model suitable for the use case by comparing transcription quality and accuracy across different models.

D. **Model Experimentation and Selection**: Various AI/ML classification models were experimented with, leveraging the transcriptions to classify the audio queries based on keywords. The performance of each model was evaluated, and the most effective model for accurately categorizing the calls was selected. Key performance metrics included precision, recall, and overall classification accuracy.

This methodology ensures a robust approach to identifying the most appropriate ASR and classification models for MFIN's needs and integration of AI into their operations.

2. Dataset

The call data is provided by MFIN.

MFIN provided 2 sets of data. First set consists of approximately 500 call recordings between the end-user and Mfin's call-center-executive. The transcription generated for these calls using ASR models were gibberish due to multiple speakers engaging at the same time.

Second set of data provided by Mfin was after-work hour calls voicemail data which is collected by the implementation of work package#1. This dataset consists of 132 after-work hour calls voicemail data.

Since this is voicemail data, and mostly contains audio of one user engaging in the conversation apart from the background noise.

MFIN provided this dataset after classifying the calls at their end. All the 132 calls details provided are classified into 5 categories:

JUNK ; NORMAL ; CRITICAL ; STAFF-RELATED ; QUERY (added later after finalizing work package)

Each call category brief description is below:

Table 1 - Distribution of data by category based on content

Category based on content	**Count**
Junk	90
Normal	34
Critical	2
Staff-related	1
Query	5
TOTAL	**132**

Additionally, Mfin provided a dictionary of words consisting of most oftenly spoken words by the end-users while reporting a complaint.

3. Models used for ASR:

We experimented with 5 popular ASR models to select the best model.

1. Google stt (Google cloud service)

2. Bhashini ASR
3. Whisper (Model available at Groq; whisper-large-v3)(API)
4. Assembly AI (API)
5. Sarvam AI (API)

4. Models used for Classification:

Due to very less and skewed data, traditional classification models will not be able to perform better. Hence we tried using the LLM Inference API provided by Groq to check the feasibility of executing our problem statement.

Models used from Groq LLM Inference API are:

- Llama3-70b-8192
- Llama-3.1-70b-versatile
- Llama-3.1-8b-instant
- Gemma2-9b-it
- Mixtral-8x7b-32768

4.1 Classification methods used

For classification task we considered using two approaches:

1. Agent based classification
2. Single prompt based classification

4.1.1 Agent based classification

We experimented with Crew AI agents to help classify our transcriptions. Created 3 agents for the complete classification task including transcribing.

- Transcriber agent: to transcribe the audio
- Grievance-officer agent: To use the transcription provided by the 'Transcriber agent' and classify the call with the help of keyword based knowledge base provided to it.
- Chief-grievance-officer agent: To supervise and recheck the Grievance-officer agent's classification. Appreciate if correctly classified, engage in discussion if incorrectly classified to get to a consensus classification category.

This approach was performing well but was taking too much time to get to the final result (avg. time taken is approx. 4 mins)

4.1.2 Single-prompt based classification

We used a single prompt which takes the transcript, instructions and the knowledge base and tries to classify the transcript. Knowledge base used here is a dictionary containing most frequently spoken words by the end-users or the callers in each category. The keyword dictionary contains words for each category. Keywords listed are in combination of both single words and in-combination words, i.e, the words can appear individually or in combination with each other.

Example: "Single Keywords": ["व्यवहार", "गाली", "मारपीट",];

”In-combination keywords”: [”रात - घर”, ”रात - फ़ोन”]

We used prompts in both Hindi and English. The prompts we used are:

The reason for using Hindi prompts is to know:

- How well the LLM is able to understand instructions in Hindi.
- Whether we can do prompt-tuning in hindi.

We also used a few English words in the Hindi prompt to check if the LLM can correlate if the transcription contains a combination of words from both languages.

Both the prompts contain the same instructions. They only differ in the language they are written in. The prompt starts with a background of the agent we are instructing it to consider as an expert in the microfinance industry who can classify the complaints into certain categories. Next we give the call transcription, the knowledge base i.e., dictionary of keywords. Then,we give it the category description for its understanding. Finally, we give the result structure which the LLM should return.

Hindi Prompt

```
prompt = f"""
आप एक विशेषज्ञ हैं जो माइक्रोफाइनेंस क्षेत्र में ग्राहक शिकायतों को वर्गीकृत करते हैं। आपको हिंदी में दी गई शिकायत का विश्लेषण करना है और उसे पांच श्रेणियों में
से एक में वर्गीकृत करना है: जंक, सामान्य, गंभीर, पूछताछ, या कर्मचारी संबंधित।

यहां ग्राहक की शिकायत का हिंदी पाठ है:
"{transcript}"

शिकायत में पाए गए कीवर्ड:
{json.dumps(found_keywords, ensure_ascii=False, indent=2)}

कृपया वर्गीकरण करते समय निम्नलिखित दिशानिर्देशों पर विचार करें:
1. Junk: यदि पाठ अर्थहीन है या समझ में नहीं आता।
2. Normal: ये नियमित मुद्दे हैं जिन्हें तत्काल बढ़ाने की आवश्यकता नहीं है।
3. Critical: इनमें गंभीर मुद्दे शामिल हैं, विशेष रूप से अशिष्ट कर्मचारियों के व्यवहार से संबंधित।
4. Query: ये सामान्य प्रश्न या जानकारी के लिए अनुरोध हैं।
5. Staff-related: इनमें विशेष रूप से कर्मचारियों या मानव संसाधन मुद्दों का उल्लेख होता है या वे शामिल होते हैं।

कृपया पाठ के संपूर्ण संदर्भ और अर्थ को ध्यान में रखते हुए इस शिकायत को वर्गीकृत करें। पाए गए कीवर्ड आपके वर्गीकरण में मदद कर सकते हैं, लेकिन वे आपके
निर्णय का एकमात्र आधार नहीं होने चाहिए। अपने वर्गीकरण के साथ एक संक्षिप्त स्पष्टीकरण और 0 से 1 के बीच एक विश्वास स्कोर प्रदान करें।

आपकी प्रतिक्रिया निम्नलिखित JSON प्रारूप में होनी चाहिए:
{{
    "classification": "Junk or Normal or Critical or Query or Staff-related",
    "explanation": "आपका स्पष्टीकरण यहाँ",
    "relevant_keywords": ["list", "of", "relevant", "keywords", "found"]
    "confidence": 0.95
}}
"""
```

English prompt:

```
prompt = f"""
Remember you are an expert in microfinance field and you classify the after-work hour audio call complaint
transcripts.
Here's the hindi transcript of the customer's complaint:
"{transcript}"

Please consider the following guidelines when classifying:
1. Junk Complaints : If the transcript is gibberish or doesn't make sense.
2. Normal Complaints: These are routine issues that don't require immediate escalation.
3. Critical Complaints: These involve serious issues, especially related to rude staff behavior.
4. Query: These are general inquiries or requests for information.
5. Staff-related: These specifically mention or involve staff members or HR issues.

Use the provided knowledge base to help with your classification. The knowledge base contains keywords and
phrases commonly associated with each category.

Knowledge Base:
{json.dumps(KNOWLEDGE_BASE, ensure_ascii=False, indent=2)}

Based on the transcript and the knowledge base, but not limited to that Knowledge Base, please classify this
complaint. Provide your classification along with a brief explanation of your reasoning.

Recheck and reason with your classification.

Your response should be in the following JSON format:
{{
    "classification": "Junk or Normal or Critical or Query or Staff-related",
    "explanation": "Your explanation here",
    "relevant_keywords": ["list", "of", "relevant", "keywords", "found"]
}}

Remember to consider the context and severity of the complaint, not just the presence of keywords.
ALWAYS PROVIDE THE CLASSIFICATION RESULT NEVER LEAVE IT EMPTY.
"""
```

5. Findings

5.1 ASR

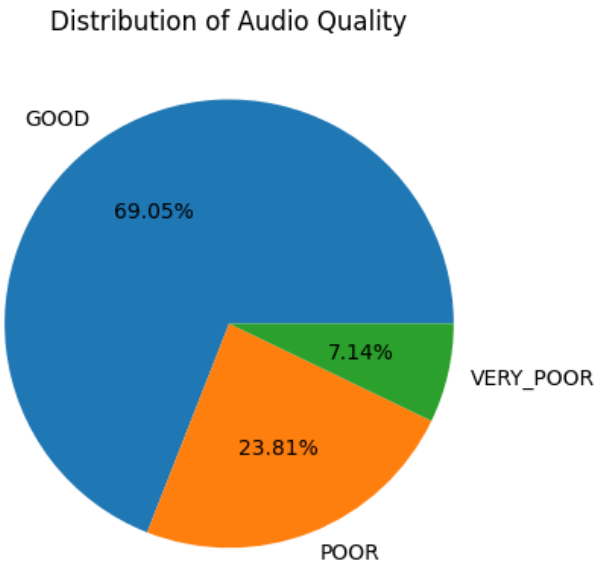
For ground truthing, we took the help of our GV moderator who listened to every single audio and transcribed it. The moderators also helped in finding the similarity between their own transcription and the ASR transcription by giving Transcription Scores.

5.1.1 Audio quality Distribution:

ASR's can only perform better if the audio quality is good. Based on this fact, the moderators helped us categorize the audio quality into 3 categories- GOOD, POOR and VERY POOR, based on the number of times the moderator has to listen to the audio to transcribe it.

- GOOD - The moderator listened to the audios for less than(<=) 2 times.
- POOR - The moderator listened to the audios for 3-5 times.
- VERY POOR - The moderator listened to the audios more than 5 times or noisy.

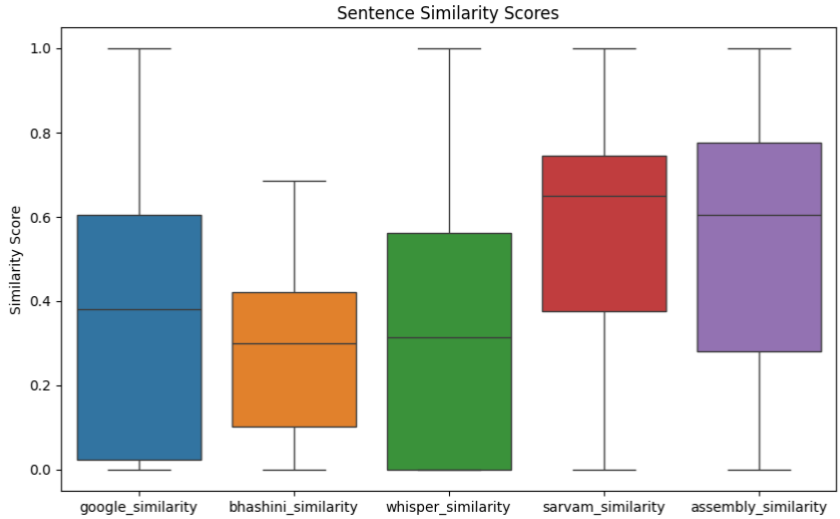
Quality	Count	Percentage
GOOD	29	69.05 %
POOR	10	23.81 %
VERY POOR	3	7.14 %



5.1.2 Quantized Similarity

For calculating similarity between moderator transcript and ASR transcript we are using TF-IDF (Term Frequency-Inverse Document Frequency) vectorization to convert the texts into numerical vectors, where each word is given a weight based on its frequency in the document and its rarity across all documents. Words that are common in a text but rare in general are given higher weights. This provides an automated, consistent way of evaluating transcription quality across all your samples.

Cosine Similarity measures the cosine of the angle between two vectors. It ranges from -1 to 1, where 1 means the vectors are identical, 0 means they're orthogonal, and -1 means they're opposites. In text analysis, it's usually between 0 and 1, where higher values indicate greater similari



From the above plot we can see that Sarvam AI transcription's similarity score w.r.t. Moderator transcription is tightly grouped.

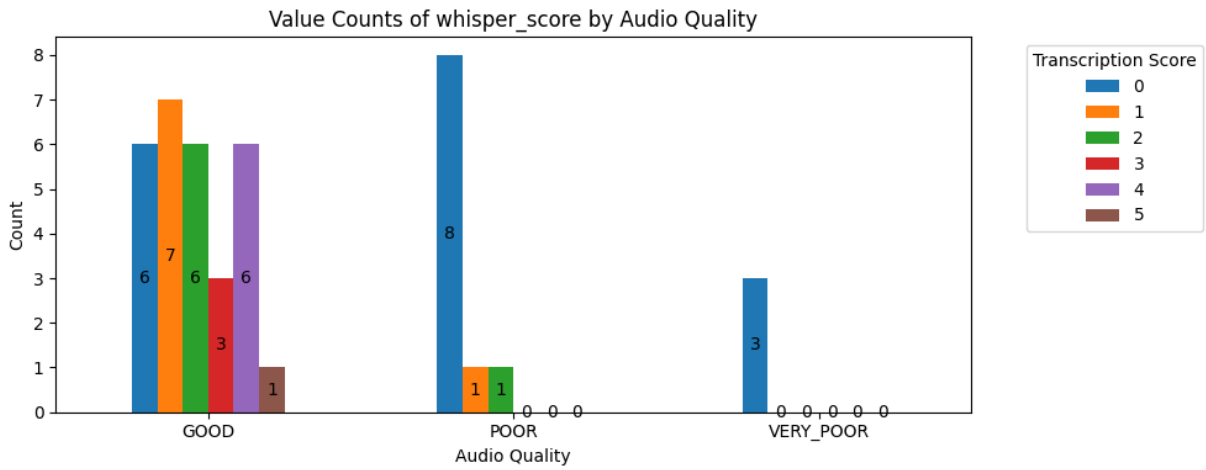
Considering the similarity scores of ASR transcription and Moderator transcription distribution plot below we are able infer that Sarvam AI and Assembly AI transcriptions outperform Whisper, Google STT and Bhashini transcriptions.

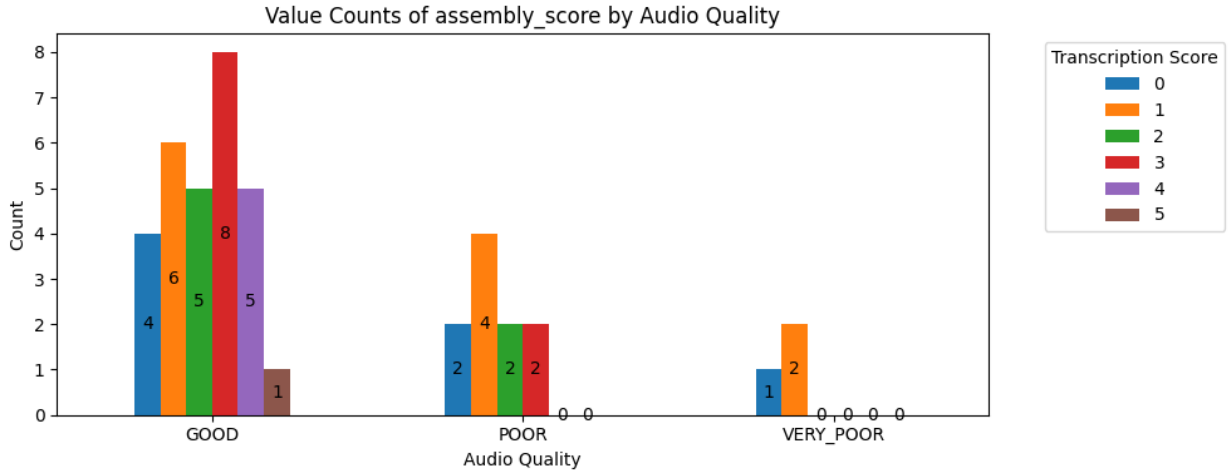
5.1.3 Transcription Score

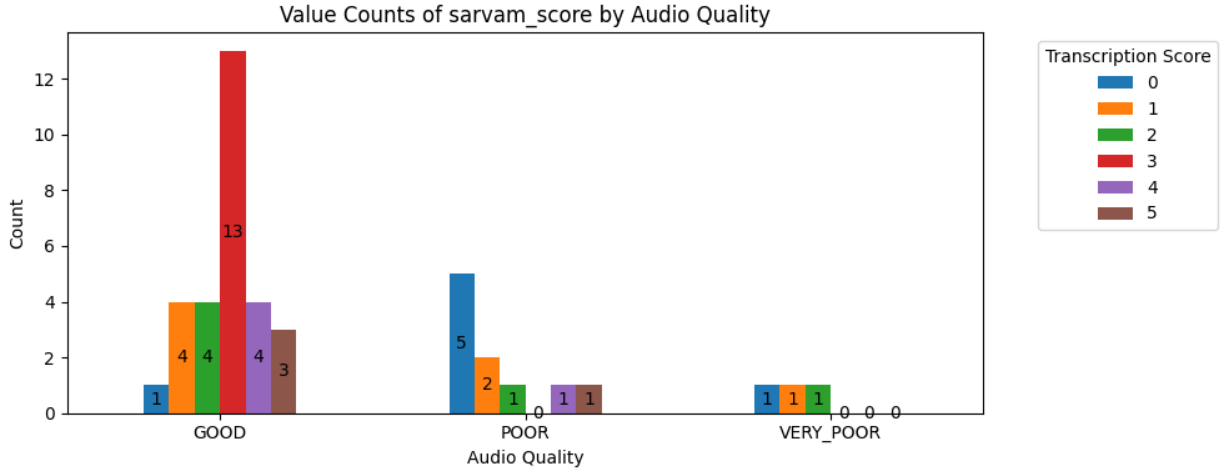
Transcription Score is a process which we at GramVaani use on a daily basis. The moderators of GV are tasked to listen and transcribe the audio. Further they have to assign a score to the automated transcription between 0-5, to check how correctly the audio is transcribed.

0 being the transcription is completely different from the audio and 5 being the transcription is exactly the same as the words spoken in the audio.

Below is the value distribution of audio quality w.r.t. Score given to the ASR transcription by the Moderator.





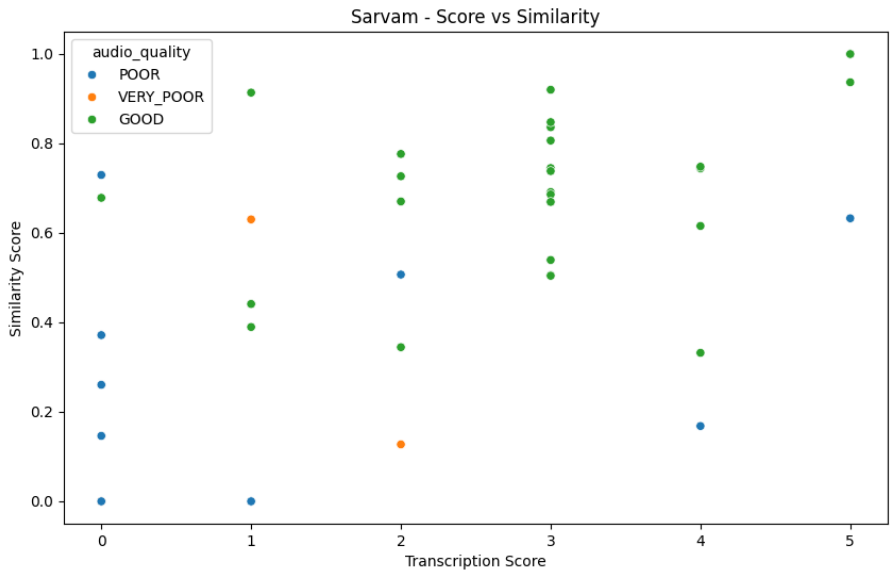


We can infer from the above plots that Sarvam AI transcription outperformed other ASR we considered.

5.1.4 Transcription Score vs Sentence-similarity

Now, we are trying to compare the performance of different transcription methods, by comparing numerically calculated scores with the manual scoring system GV had in place. It provides an automated, consistent way of evaluating transcription quality across all your samples.

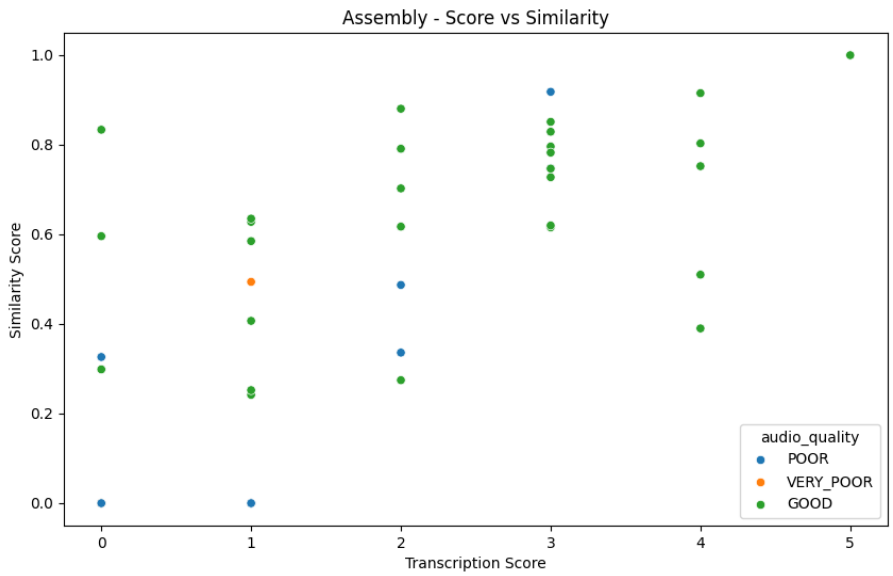
Sarvam AI



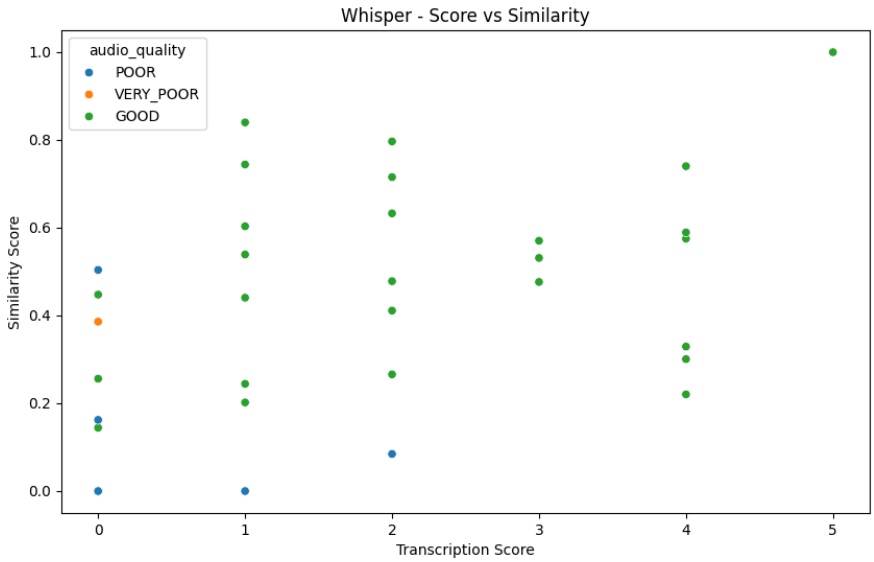
With a wide spread of performance, Sarvam AI shows a positive correlation between transcription scores and similarity scores. Good quality audio tends to have higher similarity scores, but there's significant overlap across audio qualities. There are notable outliers, such as poor quality audio with high similarity scores and good quality audio with low similarity scores. The performance seems to have a ceiling around 0.9 similarity score, even for high transcription scores.

Assembly AI

Assembly demonstrates a stronger positive correlation between transcription scores and similarity scores compared to Sarvam. There's a clearer distinction between audio quality categories, with good quality audio consistently achieving higher similarity scores. Assembly seems more robust to audio quality variations, as even some poor quality audio samples achieve high similarity scores. The spread of similarity scores narrows as transcription scores increase, indicating more consistent performance for higher-rated transcriptions.



Whisper



Whisper shows a moderate positive correlation between transcription and similarity scores, but with more variability than Assembly. Good quality audio samples are spread across a wide range of similarity scores, suggesting inconsistent performance even with good input. There are fewer data points for poor and very poor audio quality, making it difficult to draw strong conclusions about Whisper's performance on lower quality audio. The highest similarity scores are achieved only by good quality audio samples.

Aspect	Sarvam AI	Assembly AI	Whisper
Correlation	Moderate	Strong	Moderate
Impact w.r.t. Audio Quality	Moderate	Low to Moderate	High
Performance	Moderate to High	High	Low
Poor Audio Performance	Low to Moderate	Moderate	Low
Top Performance score	1.0	1.0	1.0
Best for Good Audio	Yes	Yes	Inconsistent
Low quality Audio reliability	Moderate	Moderate	Low

5.2 Classification model

We considered transcription generated by the Whisper, Assembly AI and Sarvam AI transcripts as the model transcription for the classification task ahead.

We go into each model's results for each of the transcriptions by plotting heat maps of value distribution of classification and moderator transcript classification.

The left heatmap shows the model's classification using ASR transcript, while the right shows the Moderator Transcript (MT) classification.

This comparison helps us easily identify where the model's predictions differ from the reference classification.

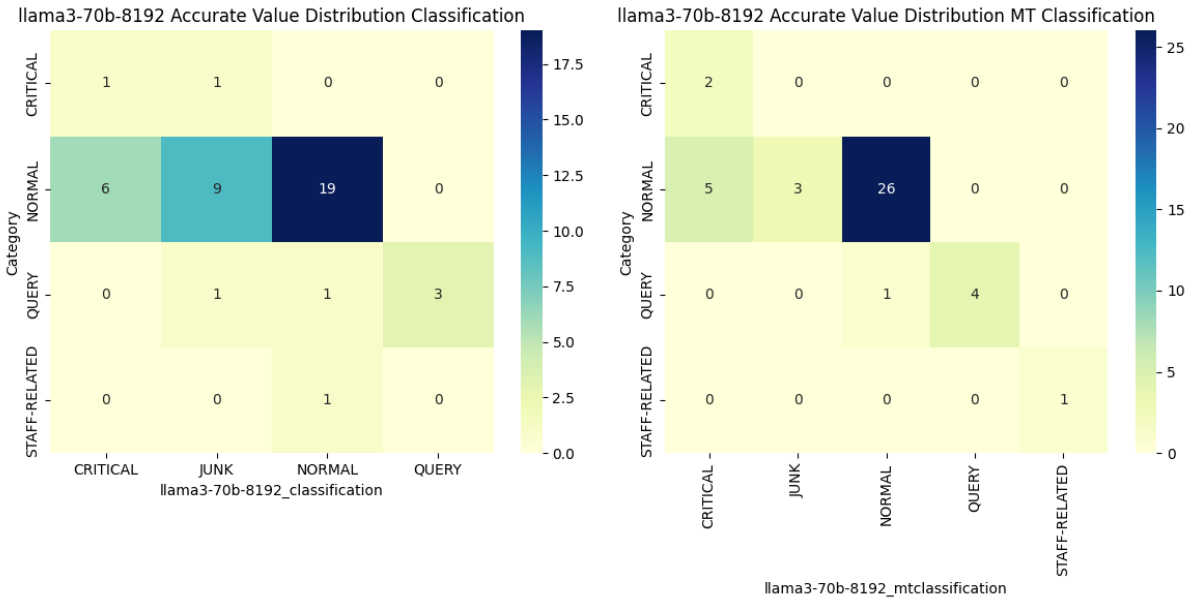
Heat maps are plotted for the distributions of model transcripts and moderator transcripts for both accurately classified and inaccurately classified transcripts. The diagonal elements in both heat maps represent correct classifications.

5.2.1 Llama3-70b-8192

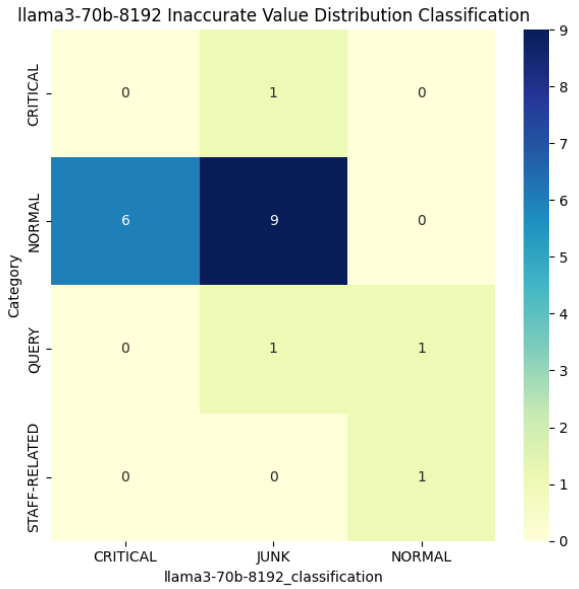
Whisper Transcript

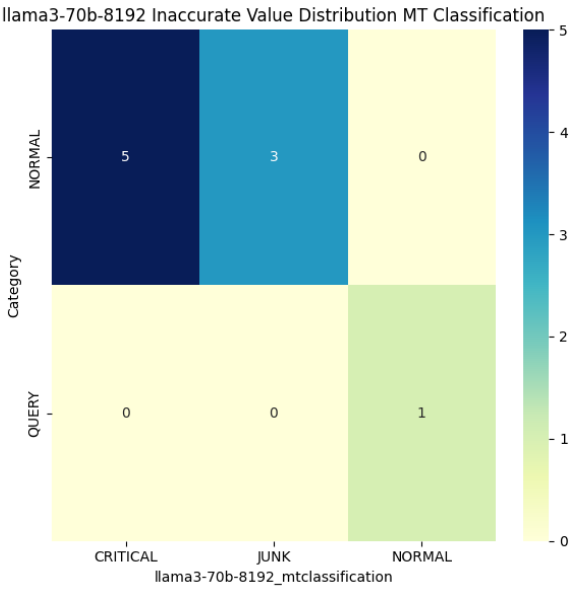
llama3-70b-8192 overall accuracy : 54.76 %

llama3-70b-8192 overall mt_accuracy : 78.57 %



- Normal Category: The model correctly classifies 19 "Normal" items, with some misclassifications as "Critical" (6) and "Junk" (9). The MT classification shows even higher accuracy for "Normal" items (26 correct).
- Critical Category: The model struggles with "Critical" items, often misclassifying them as "Normal" or "Junk". The MT classification shows better performance in identifying "Critical" items.
- Query Category: The model shows good performance in identifying "Query" items (3 correct), with some false positives. The MT classification is even more accurate for "Query" items (4 correct).
- Staff-Related Category: Both classifications rarely use this category, with only one instance in the MT classification.



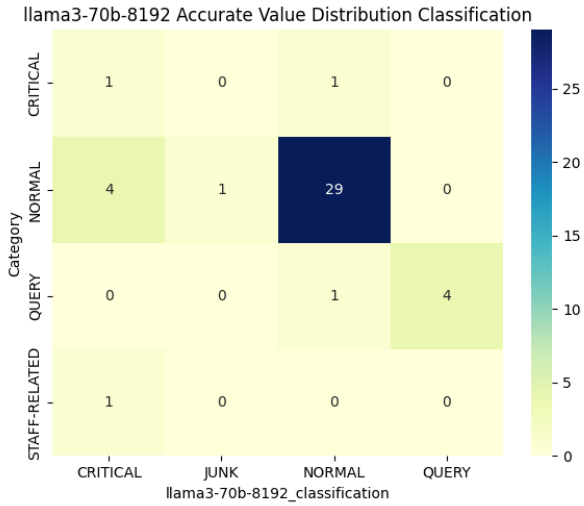


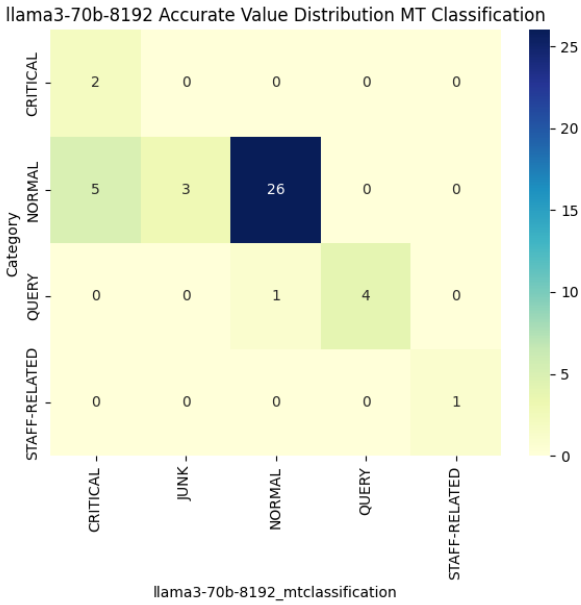
- Normal Category: In both classifications, the "Normal" category has the highest counts, suggesting it's the most common category. The model classifies more instances as "Normal" (15 total) compared to the MT classification (8 total).
- Critical Category: The model rarely classifies items as "Critical" (only 1 instance), while the MT classification has 5 "Critical" items. This suggests the model might be underestimating the criticality of some items.
- Junk Category: The model classifies 11 items as "Junk", while the MT classification has only 3. This indicates the model might be over-classifying items as "Junk".
- Query and Staff-Related Categories: These categories have low counts in both classifications, but there are some discrepancies between the model and MT classifications.

Assembly AI Transcript

llama3-70b-8192 overall accuracy : 80.95%

llama3-70b-8192 overall mt_accuracy : 78.57%



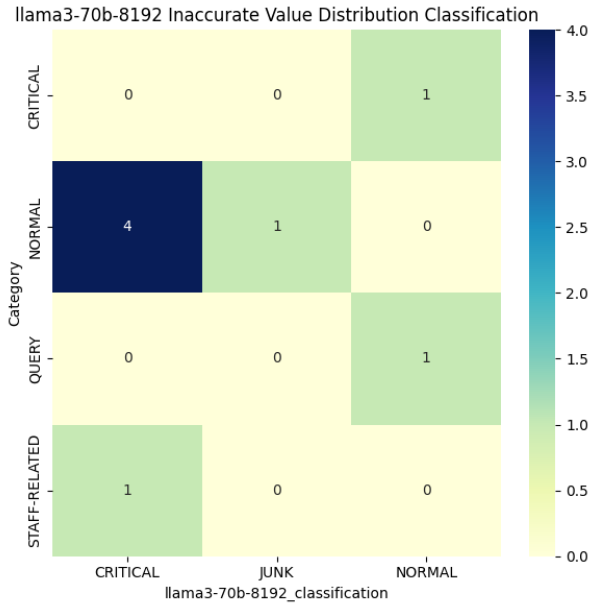


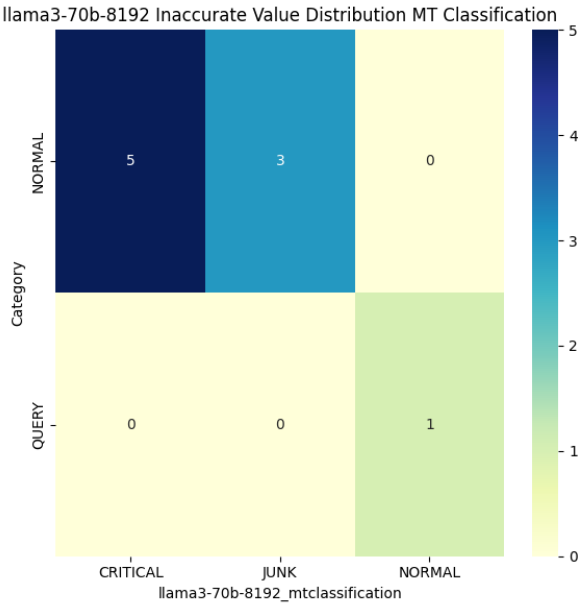
Accurate Value Distribution:

- Strong performance in classifying "Normal" items (29 correct out of 26 MT-classified).
- Correctly identifies 1 out of 2 "Critical" items.
- Perfect identification of "Query" items (4 correct out of 4).
- Under Classified "Junk" items (1 classification vs. 3 MT-classified).

Inaccurate Value Distribution:

- Overclassifies items as "Critical" (5 vs. 4 MT-classified).
- Under Classifies "Junk" items (1 vs. 3 MT-classified).
- Correctly identifies most "Normal" items (4 out of 5 MT-classified).

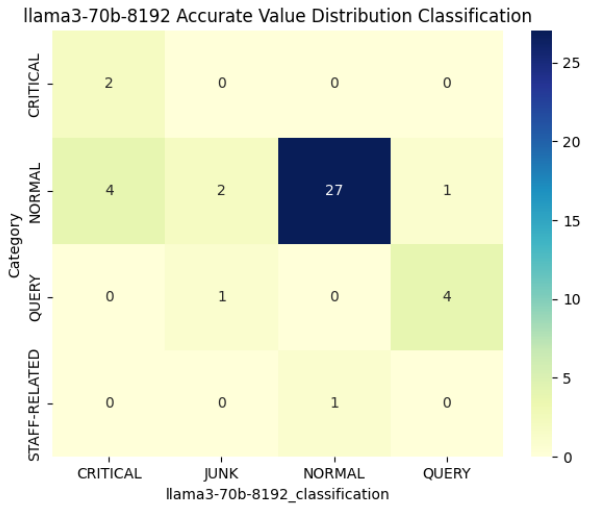


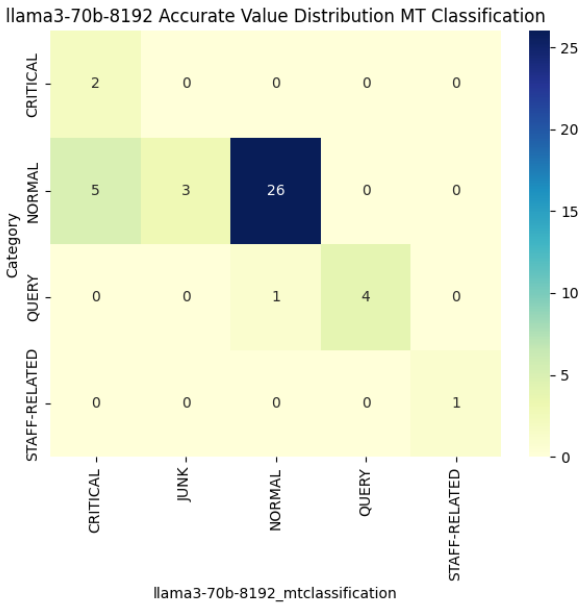


Sarvam AI Transcript

llama3-70b-8192 overall accuracy : 78.57%

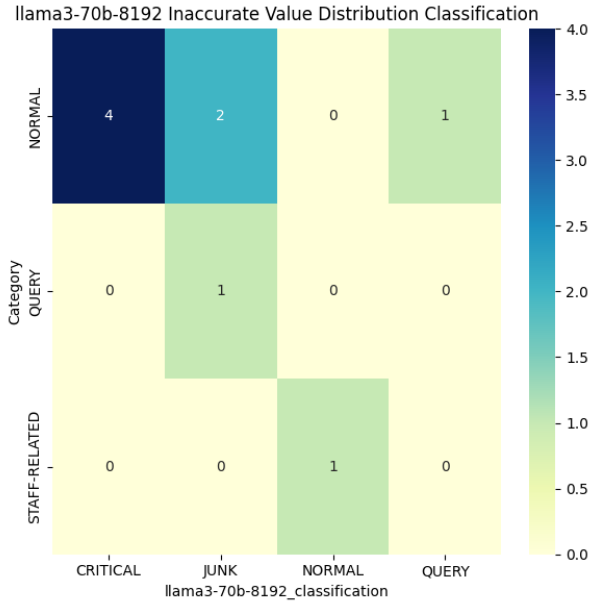
llama3-70b-8192 overall mt_accuracy : 78.57%

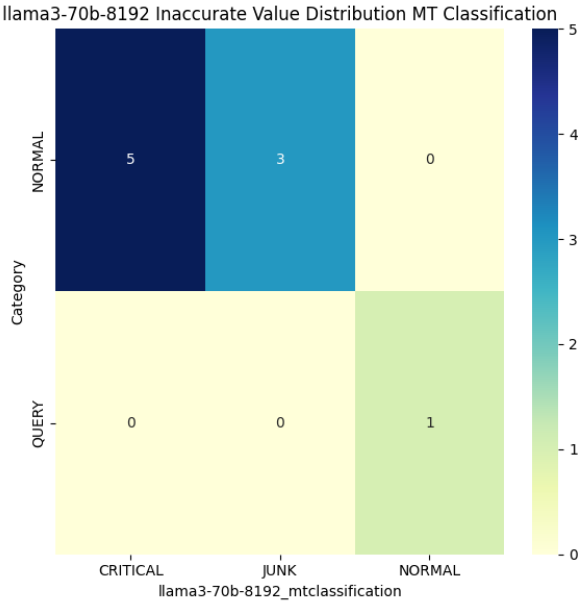




Accurate Value Distribution:

- Normal Classification: Excellent performance, correctly identifying 27 out of 26 MT-classified Normal items.
- Critical Classification: The model correctly identifies both MT-classified Critical items.
- Junk Classification: The model classifies 2 items as Junk, while MT classifies 3.
- Query Classification: Perfect performance, correctly identifying all 4 MT-classified Query items.
- Staff-Related Classification: The model correctly identifies the single MT-classified Staff-Related item.





Inaccurate Value Distribution:

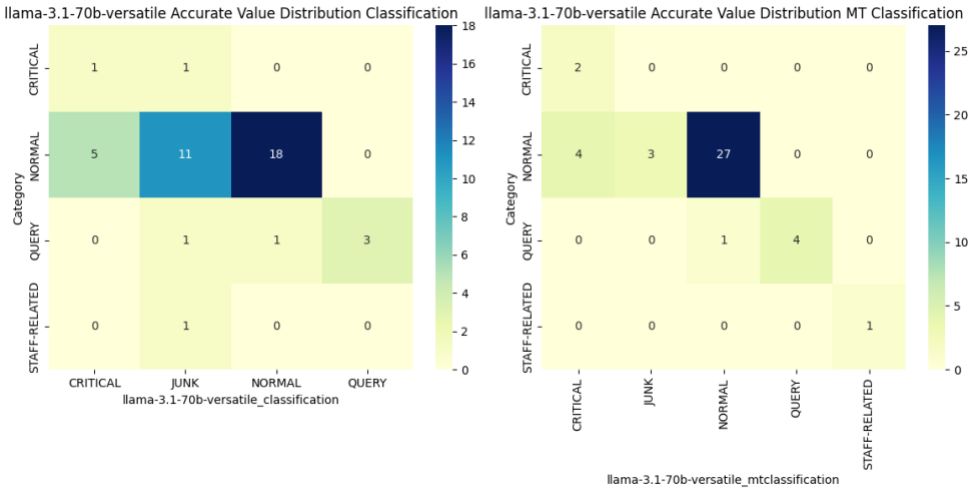
- Normal Classification: The model correctly identifies 6 out of 8 MT-classified Normal items.
- Critical Classification: The model correctly identifies 4 out of 5 MT-classified Critical items.
- Junk Classification: The model classifies 2 items as Junk, while MT classifies 3.
- Query Classification: The model correctly identifies 1 Query item.
- Staff-Related Classification: The model correctly identifies 1 Staff-Related item.

5.2.2 Llama-3.1-70b-versatile

Whisper Transcript

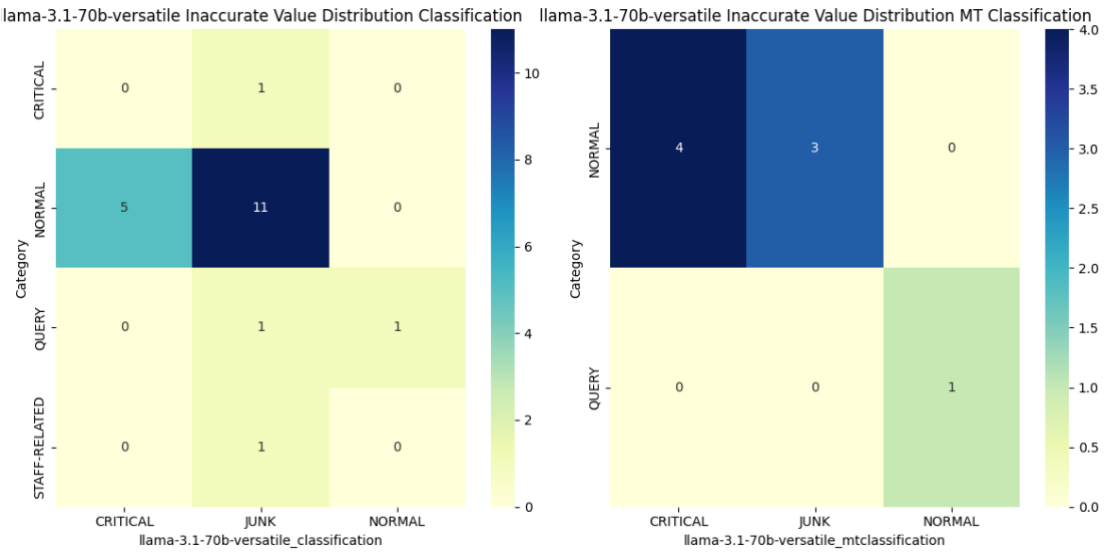
llama-3.1-70b-versatile overall accuracy : 52.38%

llama-3.1-70b-versatile overall mt_accuracy : 80.95%



Accurate Value Distribution:

- The model performs very well in classifying "Normal" items, with 18 correct classifications.
- The model struggles with "Critical" items, only correctly identifying 1 out of 2.
- The model shows good performance in identifying "Query" items (3 correct), slightly less than MT (4 correct).
- Both classifications rarely use the "Staff-Related" category.



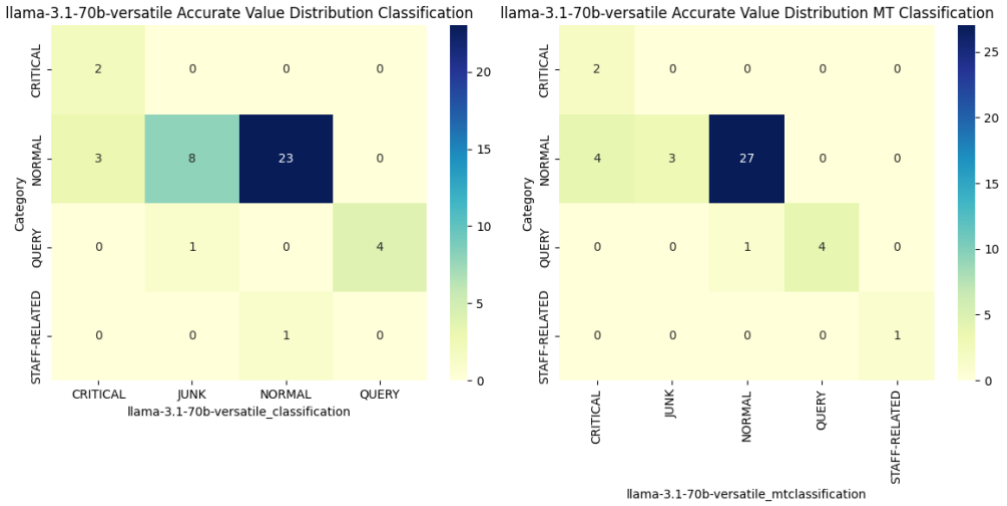
Inaccurate Value Distribution:

- The model heavily favors "Normal" and "Junk" classifications, while MT shows a more balanced distribution.
- The model fails to identify any "Critical" items, while MT classifies 4 items as "Critical".
- The model classifies more items as "Junk" (14) compared to MT (3).
- MT classification shows a more even distribution between "Normal" and "Junk" categories.

Assembly AI Transcript

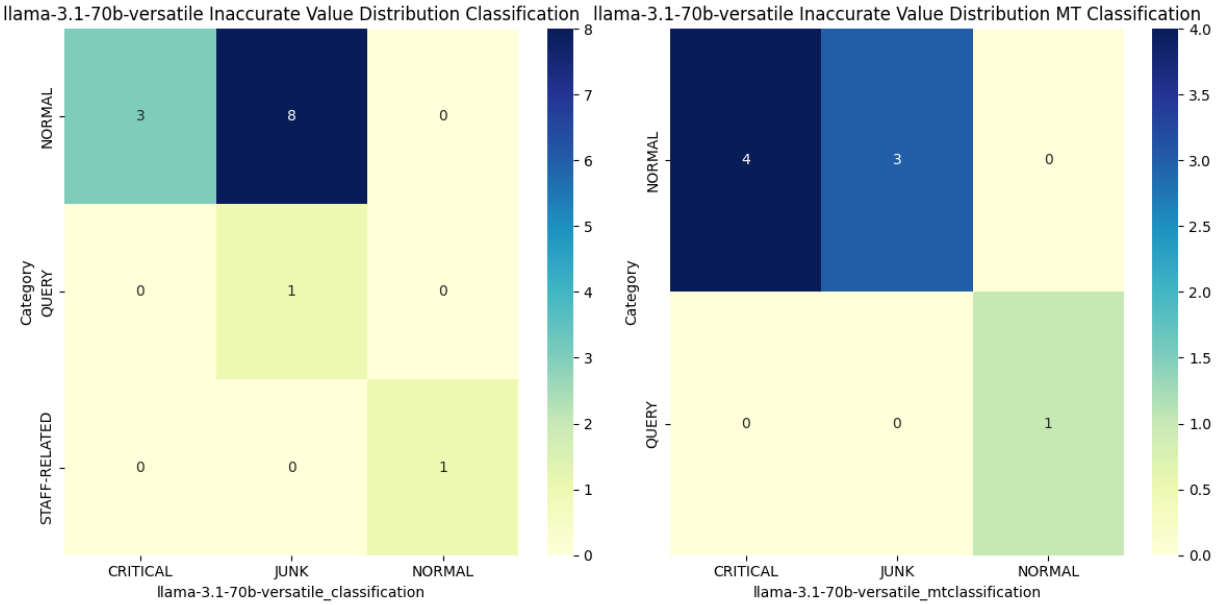
llama-3.1-70b-versatile overall accuracy : 69.04%

llama-3.1-70b-versatile overall mt_accuracy : 80.95%



Accurate Value Distribution:

- Excellent performance in classifying "Normal" items (23 correct out of 27 MT-classified).
- Perfect identification of "Critical" items (2 out of 2).
- Good at identifying "Query" items (4 correct out of 4).
- Tendency to over classify items as "Junk" (8 classifications vs. 3 MT-classified).



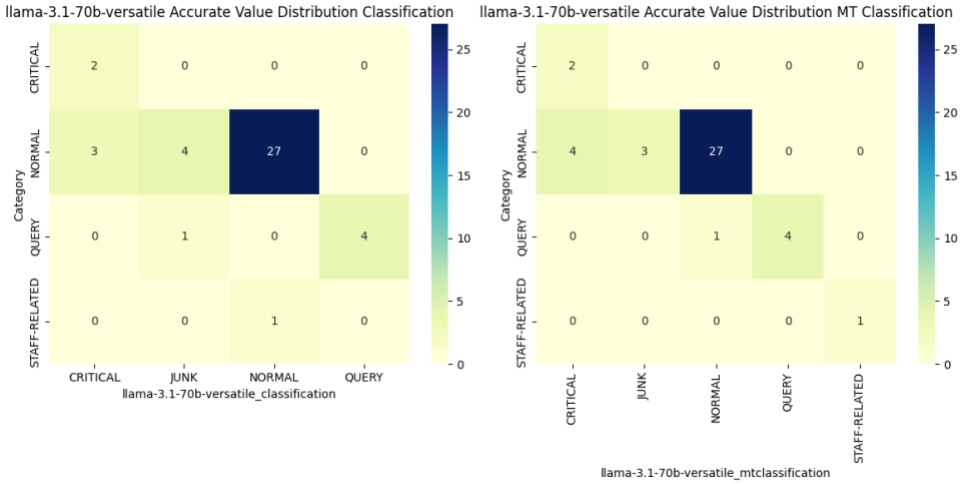
Inaccurate Value Distribution:

- Overclassifies items as "Junk" (9 vs. 3 MT-classified).
- Under Classifies "Critical" items (3 vs. 4 MT-classified).
- Correctly identifies all "Normal" items (8 out of 7 MT-classified).

Sarvam AI Transcript

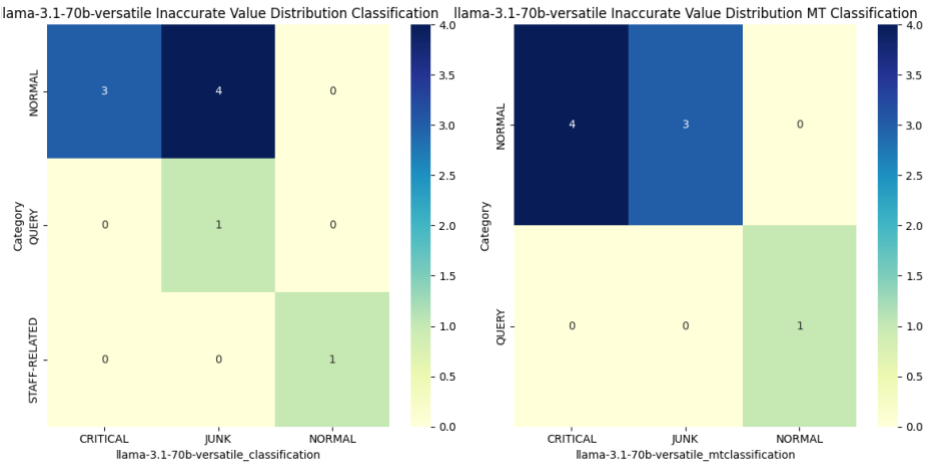
llama-3.1-70b-versatile overall accuracy : 78.57%

llama-3.1-70b-versatile overall mt_accuracy : 80.95%



Accurate Value Distribution:

- Normal Classification: Excellent performance, correctly identifying 27 out of 27 MT-classified Normal items.
- Critical Classification: The model correctly identifies both MT-classified Critical items.
- Junk Classification: The model classifies 4 items as Junk, while MT classifies 3.
- Query Classification: Perfect performance, correctly identifying all 4 MT-classified Query items.
- Staff-Related Classification: The model correctly identifies the single MT-classified Staff-Related item.



Inaccurate Value Distribution:

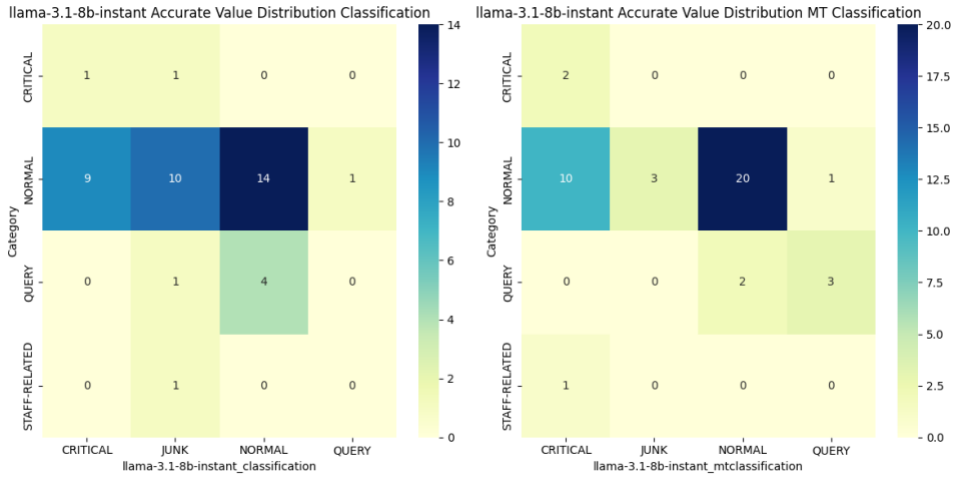
- Normal Classification: The model correctly identifies all 7 MT-classified Normal items.
- Critical Classification: The model misclassifies 3 Critical items as Normal.
- Junk Classification: The model classifies 4 items as Junk, while MT classifies 3.
- Query Classification: The model correctly identifies 1 Query item.
- Staff-Related Classification: The model correctly identifies 1 Staff-Related item.

5.2.3 Llama-3.1-8b-instant

Whisper Transcript

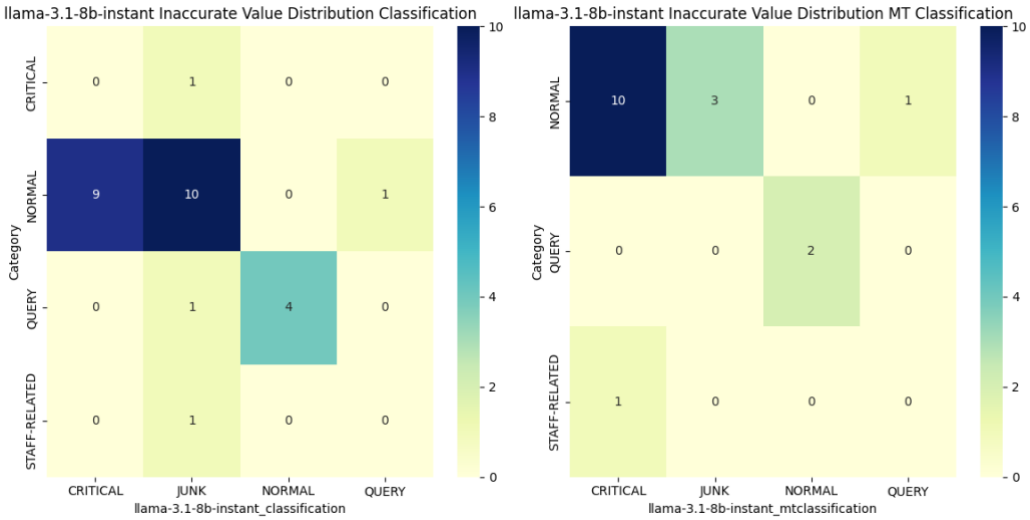
llama-3.1-8b-instant overall accuracy : 35.71%

llama-3.1-8b-instant overall mt_accuracy : 59.52%



Accurate Value Distribution:

- The model performs well in classifying "Normal" items, with 14 correct classifications.
- The model struggles with "Critical" items, only correctly identifying 1 out of 2.
- The model shows good performance in identifying "Query" items (4 correct), slightly better than MT (3 correct).
- Both classifications rarely use the "Staff-Related" category.



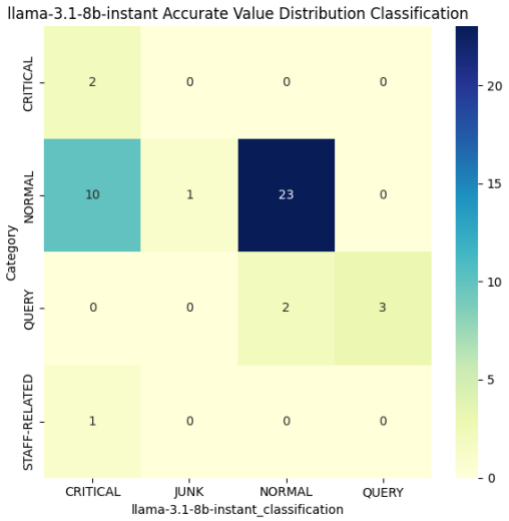
Inaccurate Value Distribution:

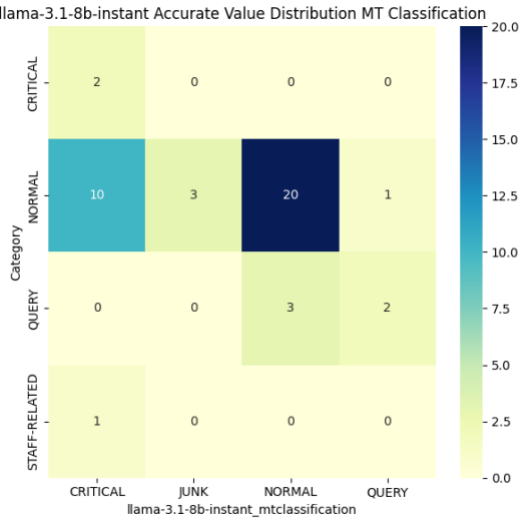
- The model tends to over classify items as "Normal" and "Junk" compared to the MT classification.
- The model fails to identify "Critical" items, while MT classifies 10 items as "Critical".
- The model identifies more "Query" items (4) than MT (2).
- MT classification shows more diverse distribution across categories.

Assembly AI Transcript

llama-3.1-8b-instant overall accuracy : 66.66%

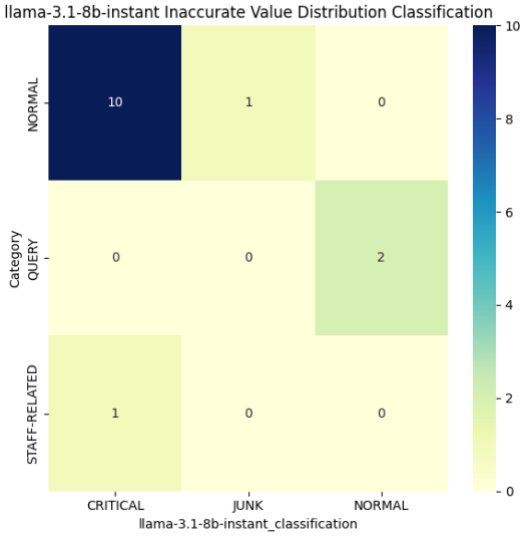
llama-3.1-8b-instant overall mt_accuracy : 57.14%

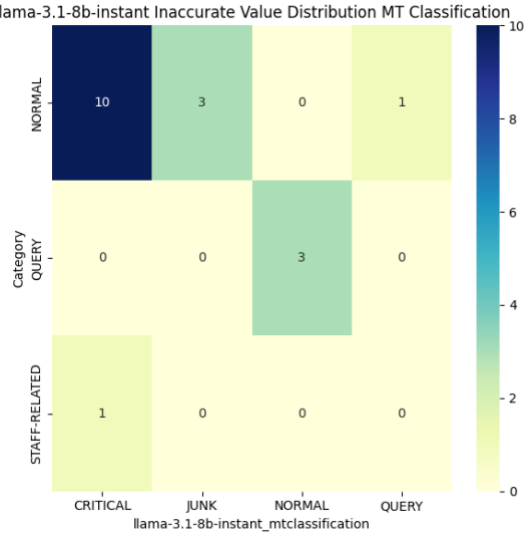




Accurate Value Distribution:

- Excellent performance in classifying "Normal" items (23 correct out of 20 MT-classified).
- Perfect identification of "Critical" items (2 out of 2).
- Good at identifying "Query" items (3 correct out of 2 MT-classified).
- Under Classified "Junk" items (1 classification vs. 3 MT-classified).





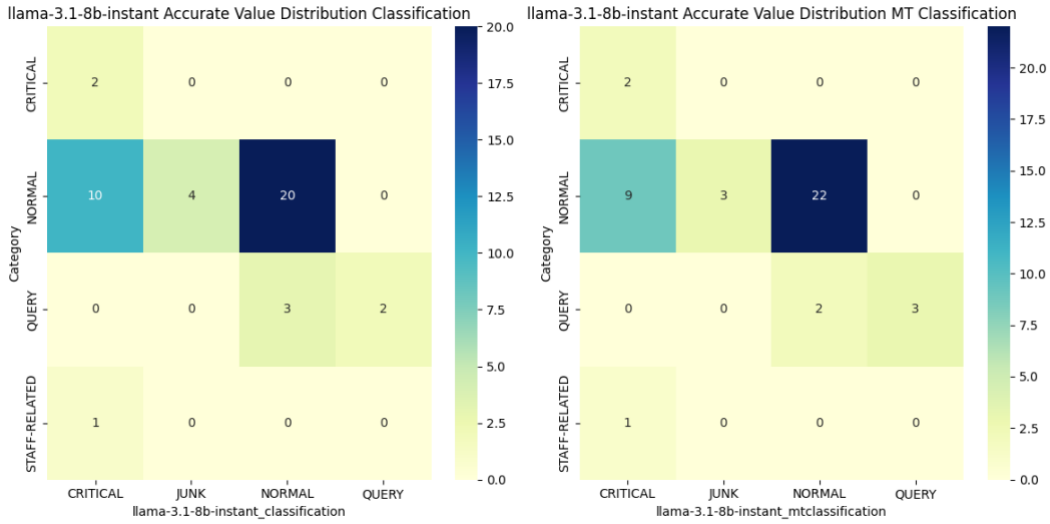
Inaccurate Value Distribution:

- Overclassifies items as "Critical" (11 vs. 10 MT-classified).
- Under Classifies "Junk" items (1 vs. 3 MT-classified).
- Correctly identifies most "Normal" items (11 out of 14 MT-classified).
- Under Classifies "Query" items (2 vs. 3 MT-classified).

Sarvam AI Transcript

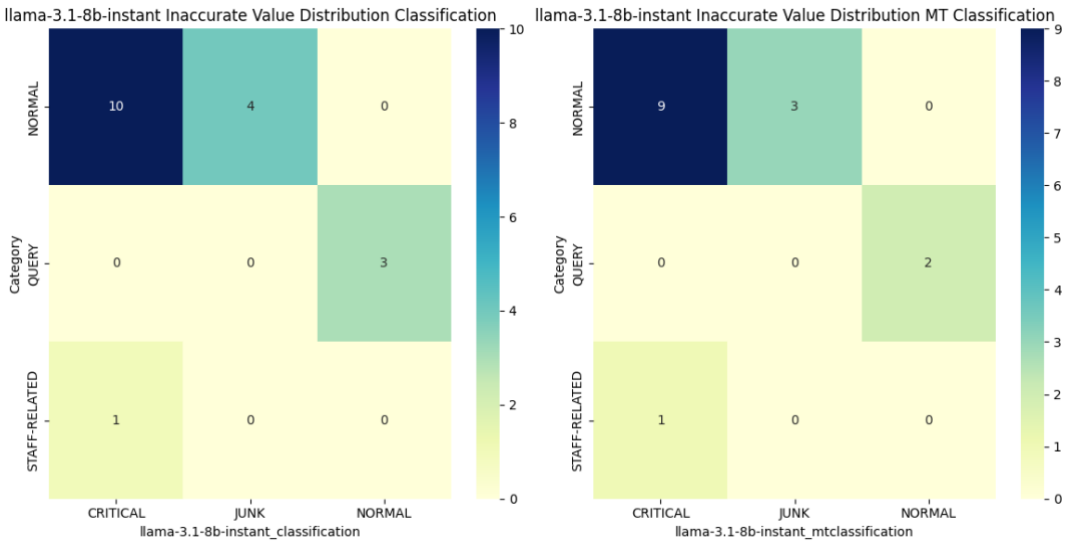
llama-3.1-8b-instant overall accuracy : 57.14%

llama-3.1-8b-instant overall mt_accuracy : 64.28%



Accurate Value Distribution:

- Normal Classification: Excellent performance, correctly identifying 20 out of 22 MT-classified Normal items.
- Critical Classification: The model correctly identifies both MT-classified Critical items.
- Junk Classification: The model slightly over classifies Junk items (4 vs. 3 MT-classified).
- Query Classification: Good performance, correctly identifying 2 out of 3 MT-classified Query items.
- Staff-Related Classification: The model correctly identifies the single MT-classified Staff-Related item.



Inaccurate Value Distribution:

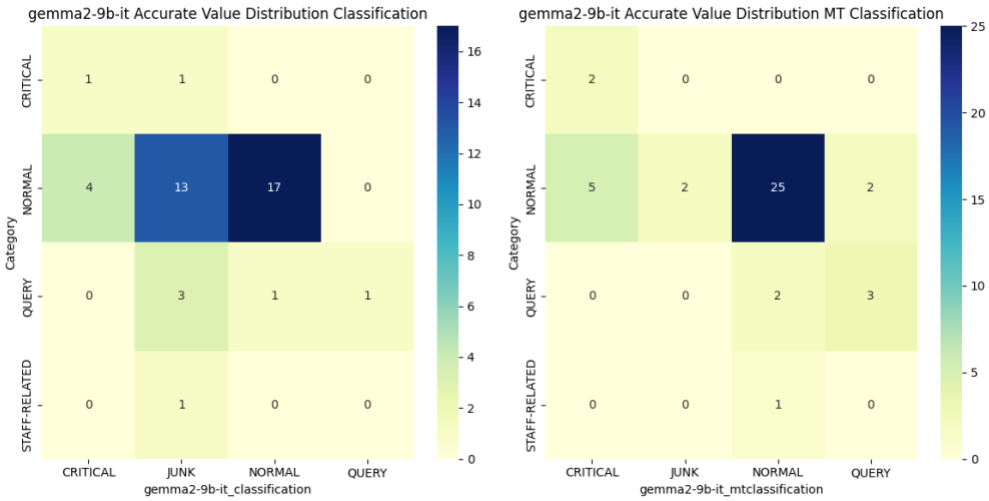
- Normal Classification: The model correctly identifies 12 out of 12 MT-classified Normal items.
- Critical Classification: The model correctly identifies 10 out of 9 MT-classified Critical items.
- Junk Classification: The model classifies 4 items as Junk, while MT classifies 3.
- Query Classification: The model over classifies Query items (3 vs. 2 MT-classified).
- Staff-Related Classification: The model correctly identifies 1 Staff-Related item.

5.2.4 Gemma2-9b-it

Whisper Transcript

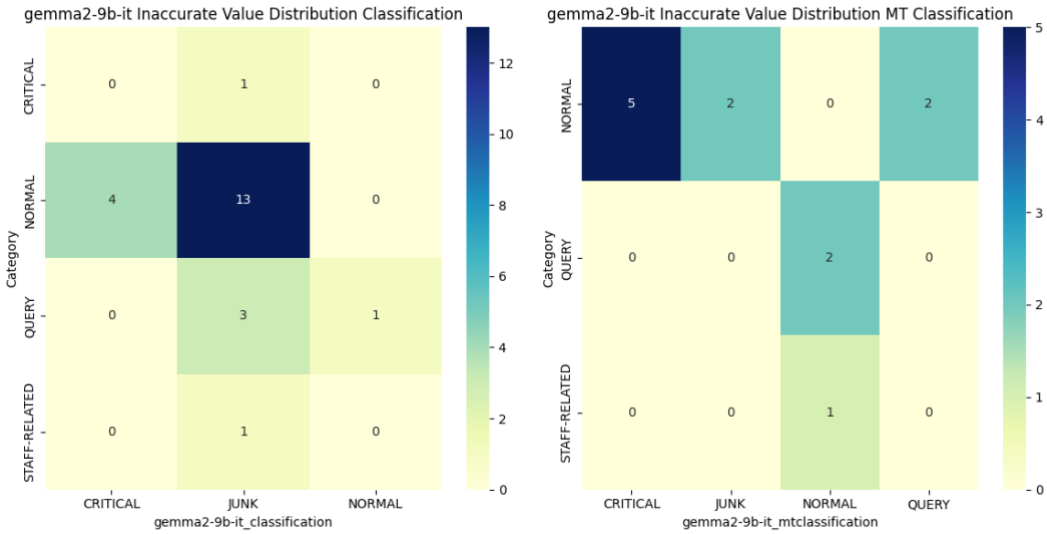
gemma2-9b-it overall accuracy : 45.23809523809524

gemma2-9b-it overall mt_accuracy : 71.42857142857143



Accurate Value Distribution:

- Strong performance in classifying "Normal" items (17 correct out of 25 MT-classified).
- Struggles with "Critical" items, correctly identifying only 1 out of 2.
- Good at identifying "Query" items (1 correct out of 3, with 3 false positives).
- Tendency to over classify items as "Junk" (13 classifications vs. 2 MT-classified).



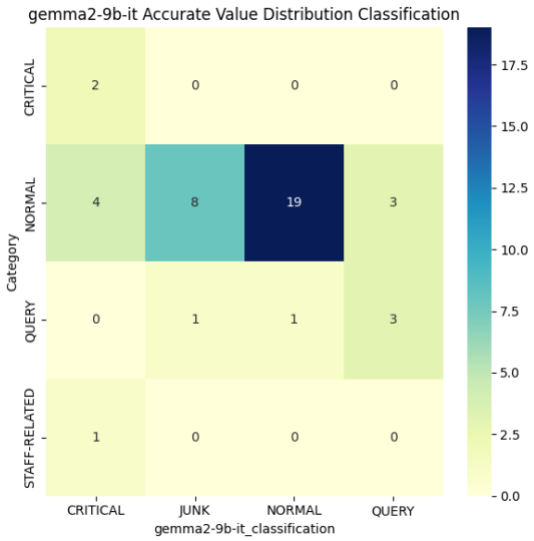
Inaccurate Value Distribution:

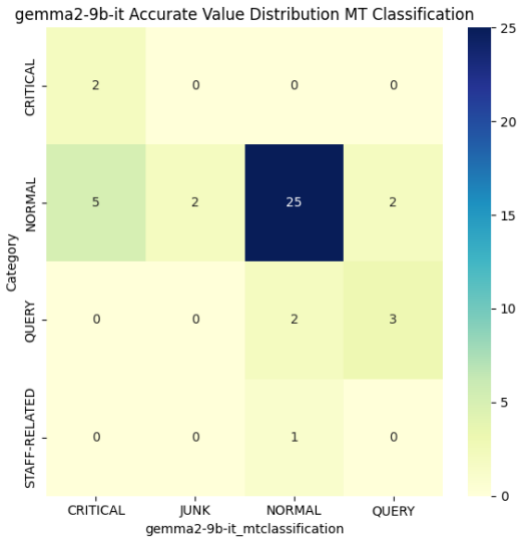
- Overclassifies items as "Normal" (13 vs. 9 MT-classified) and "Junk" (18 vs. 2 MT-classified).
- Fails to identify any "Critical" items (0 vs. 5 MT-classified).
- Slightly better at identifying "Query" items compared to MT classification.

Assembly AI Transcript

gemma2-9b-it overall accuracy : 57.14%

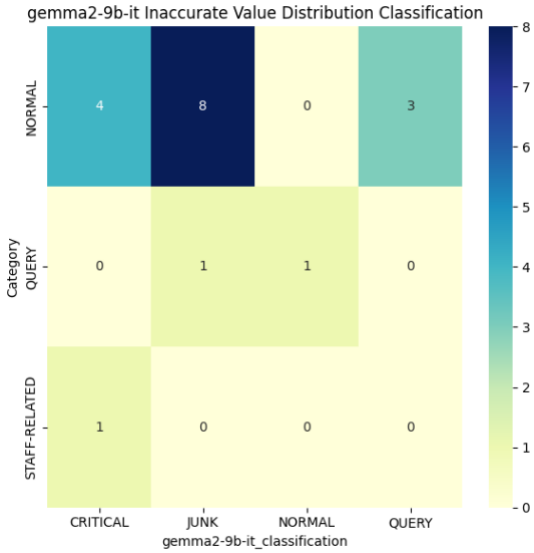
gemma2-9b-it overall mt_accuracy : 71.42%

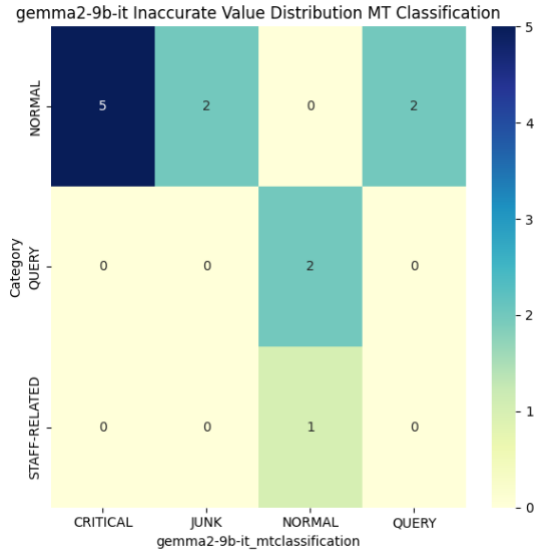




Accurate Value Distribution:

- Strong performance in classifying "Normal" items (19 correct out of 25 MT-classified).
- Perfect identification of "Critical" items (2 out of 2).
- Good at identifying "Query" items (3 correct out of 3).
- Overclassifies items as "Junk" (8 classifications vs. 2 MT-classified).





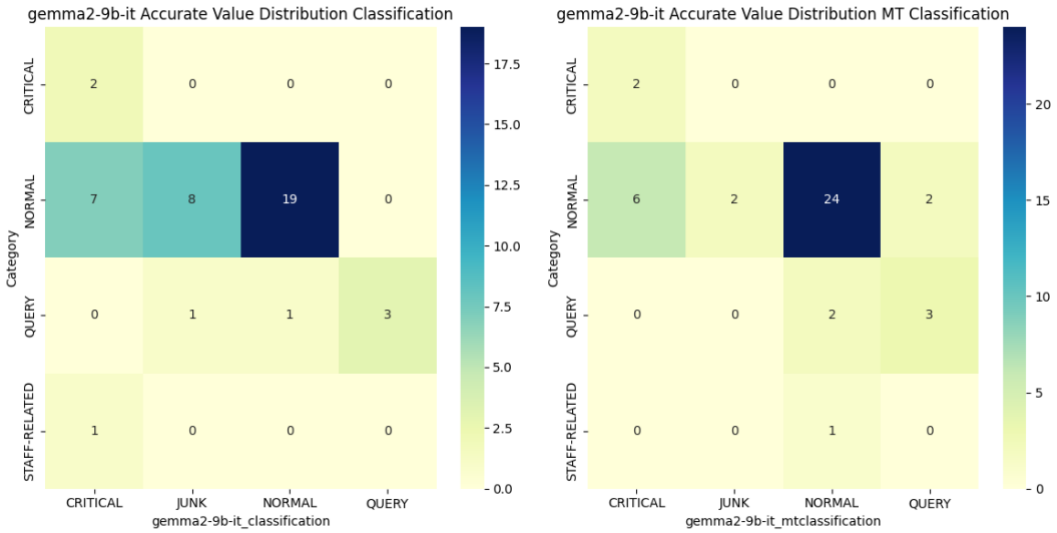
Inaccurate Value Distribution:

- Overclassifies items as "Junk" (8 vs. 2 MT-classified).
- Under Classifies "Critical" items (4 vs. 5 MT-classified).
- Correctly identifies most "Normal" items (11 out of 9 MT-classified).
- Misses "Query" items (0 vs. 2 MT-classified).

Sarvam AI Transcript

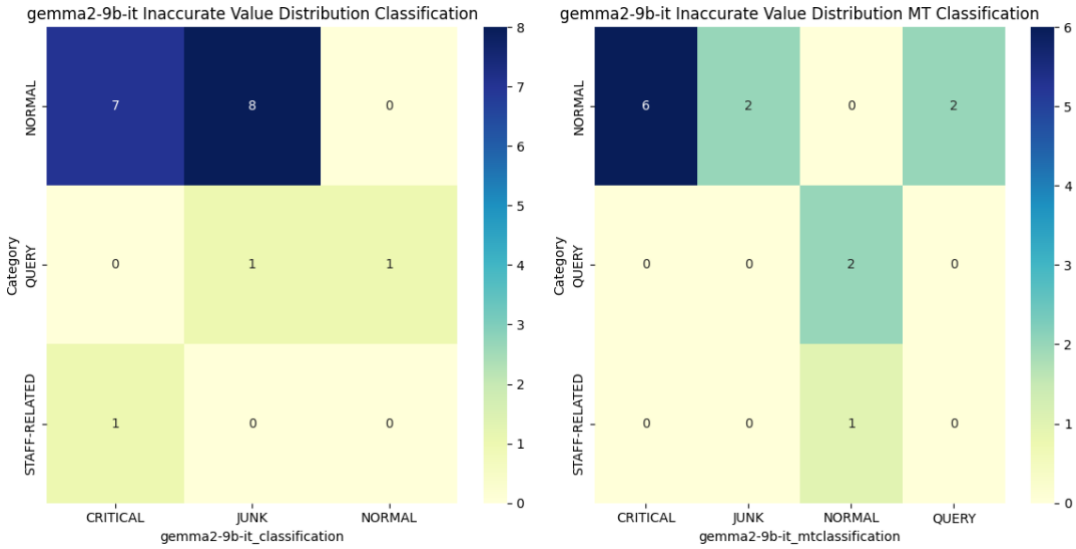
gemma2-9b-it overall accuracy : 57.14%

gemma2-9b-it overall mt_accuracy : 69.04%



Accurate Value Distribution:

- Normal Classification: Good performance, correctly identifying 19 out of 24 MT-classified Normal items.
- Critical Classification: The model correctly identifies both MT-classified Critical items.
- Junk Classification: The model over classifies Junk items (9 vs. 2 MT-classified).
- Query Classification: Good performance, correctly identifying 3 out of 5 MT-classified Query items.
- Staff-Related Classification: The model correctly identifies the single MT-classified Staff-Related item.



Inaccurate Value Distribution:

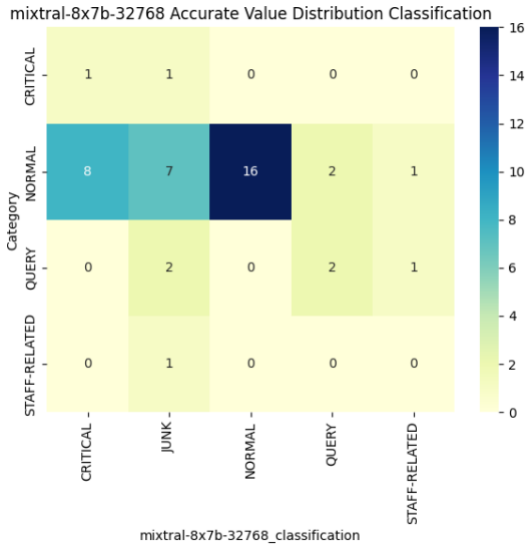
1. Normal Classification: The model over classifies Normal items (15 vs. 10 MT-classified).
2. Critical Classification: The model misclassified all 6 Critical items as Normal or Junk.
3. Junk Classification: The model over classifies Junk items (9 vs. 2 MT-classified).
4. Query Classification: The model identifies 2 Query items, matching MT classification.
5. Staff-Related Classification: The model correctly identifies 1 Staff-Related item.

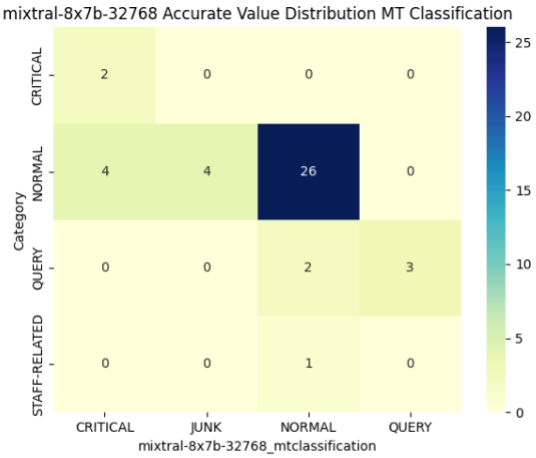
5.2.5 Mixtral-8x7b-32768

Whisper Transcript

mixtral-8x7b-32768 overall accuracy : 45.24%

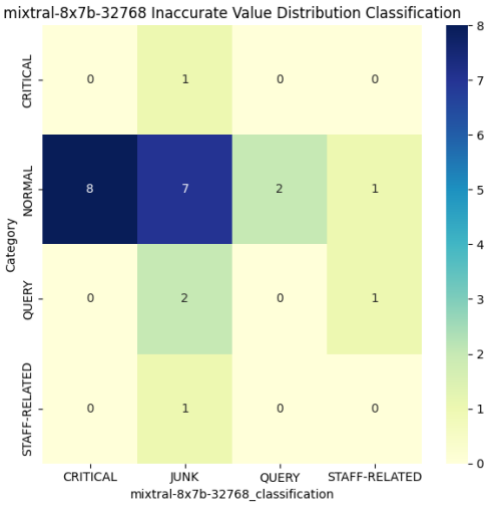
mixtral-8x7b-32768 overall mt_accuracy : 73.81%

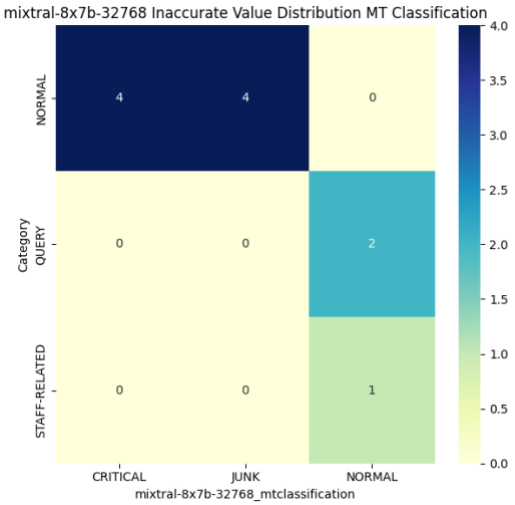




Accurate Value Distribution:

- Excellent performance in classifying "Normal" items (16 correct out of 26 MT-classified).
- Struggles with "Critical" items, correctly identifying only 1 out of 2.
- Good at identifying "Query" items (2 correct out of 3, with 2 false positives).
- Tendency to over classify items as "Junk" (7 classifications vs. 4 MT-classified).





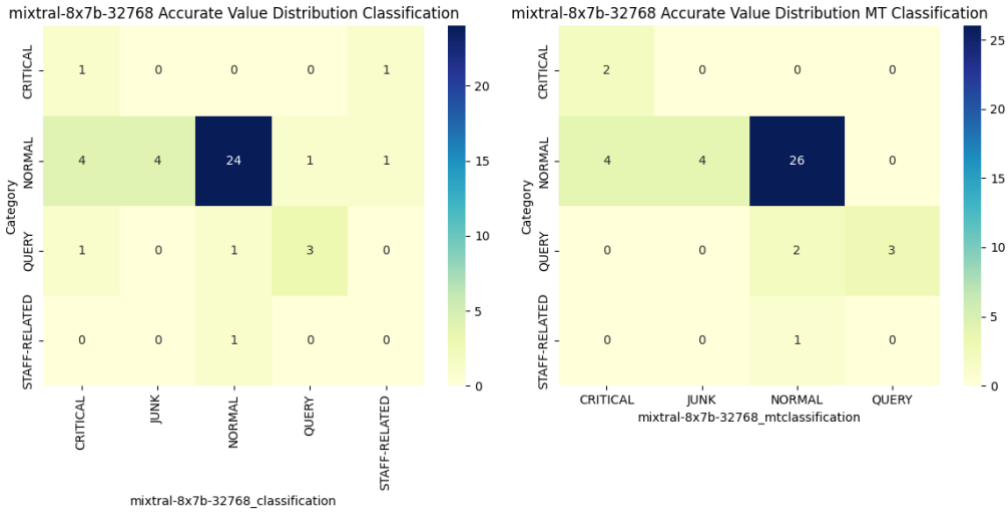
Inaccurate Value Distribution:

- Overclassifies items as "Normal" (17 vs. 8 MT-classified) and "Critical" (8 vs. 4 MT-classified).
- Accurately identifies "Junk" items (7 vs. 4 MT-classified).
- Slightly better at identifying "Query" items compared to MT classification.
- Introduces a "Staff-Related" category not present in MT classification.

Assembly AI Transcript

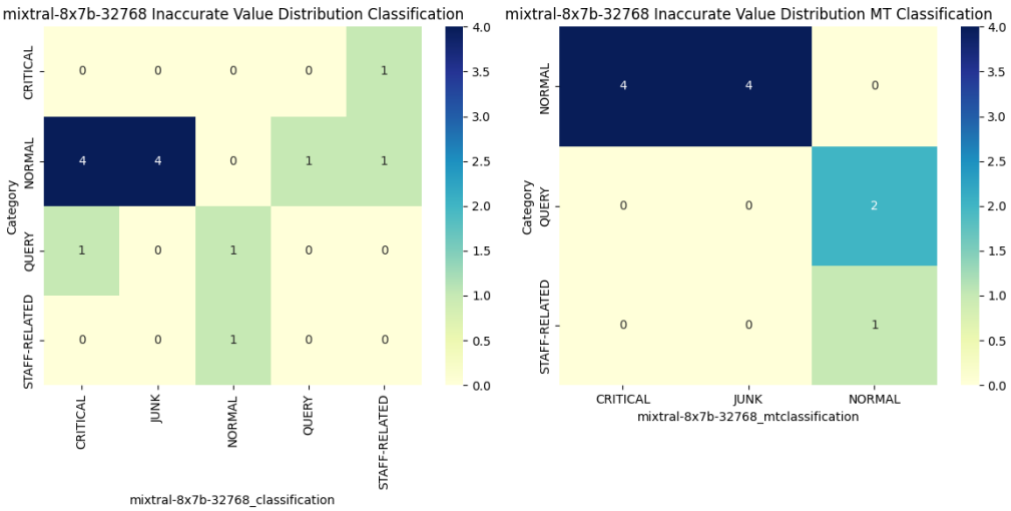
mixtral-8x7b-32768 overall accuracy : 66.66%

mixtral-8x7b-32768 overall mt_accuracy : 73.81%



Accurate Value Distribution:

- Normal Classification: Excellent performance, correctly identifying 24 out of 26 MT-classified Normal items.
- Critical Classification: The model correctly identifies 1 out of 2 MT-classified Critical items.
- Junk Classification: The model classifies 4 items as Junk, matching the MT classification.
- Query Classification: Good performance, correctly identifying 3 out of 3 MT-classified Query items.
- Staff-Related Classification: The model correctly identifies the single MT-classified Staff-Related item.



Inaccurate Value Distribution:

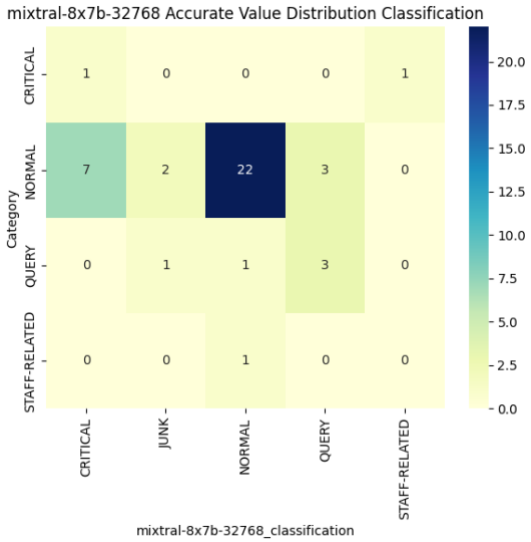
- Normal Classification: The model correctly identifies 4 out of 8 MT-classified Normal items, showing moderate accuracy.
- Critical Classification: The model fails to identify any Critical items, while MT classifies 4 as Critical.
- Junk Classification: The model classifies 4 items as Junk, matching the MT classification.
- Query Classification: The model identifies 1 Query item, while MT classifies 2.

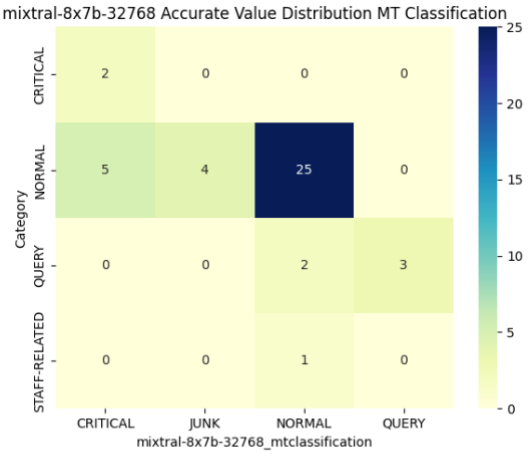
- Staff-Related Classification: The model correctly identifies 1 Staff-Related item, matching the MT classification.

Sarvam AI Transcript

mixtral-8x7b-32768 overall accuracy : 61.91%

mixtral-8x7b-32768 overall mt_accuracy : 71.43%





Accurate Value Distribution:

- Normal Classification: Excellent performance, correctly identifying 22 out of 25 MT-classified Normal items.
- Critical Classification: The model correctly identifies 1 out of 2 MT-classified Critical items.
- Junk Classification: The model classifies 2 items as Junk, while MT classifies 4.
- Query Classification: Good performance, correctly identifying 3 out of 5 MT-classified Query items.
- Staff-Related Classification: The model correctly identifies the single MT-classified Staff-Related item.

Inaccurate Value Distribution:

1. Normal Classification: The model correctly identifies 7 out of 9 MT-classified Normal items.
2. Critical Classification: The model fails to identify any Critical items, while MT classifies 5.
3. Junk Classification: The model classifies 2 items as Junk, while MT classifies 4.
4. Query Classification: The model identifies 2 Query items, matching MT classification.
5. Staff-Related Classification: The model correctly identifies 1 Staff-Related item, matching MT classification.

6. Observations

6.1 ASR

- As per our experimentations the best among the 5 ASR models are ***Sarvam AI and Assembly AI***. Even though Assembly AI scores more accuracy, Sarvam AI has its similarity scores is tightly packed than Assembly AI.

6.2 Classification task

Consider the below mentioned scores:

- Most accurate Model is Llama3-70b-8192 with accuracy score of 80.95% using Assembly Transcription and 78.57% accuracy with Sarvam AI transcription.
- Least accurate Model is Llama-3.1-8b-instant.

Model	Whisper accuracy	Whisper mtaccuracy	Assembly accuracy	Assembly mtaccuracy	Sarvam accuracy	Sarvam mtaccuracy
Llama3-70b-8192	54.76%	78.57%	***80.95%***	78.57%	*78.57%*	*78.57%*
Llama-3.1-70b-versatile	52.38%	80.95%	69.04%	80.95%	*78.57%*	*80.95%*
Llama-3.1-8b-instant	35.71%	59.52%	66.66%	57.14%	57.14%	64.28%
Gemma2-9b-it	45.23%	71.42%	57.14%	71.42%	57.14%	69.04%
Mixtral-8x7b-32768	45.23%	73.80%	66.66%	73.80%	61.90%	71.42%

6.3 Observations during result verification

- It has been observed that due to some semantic differences the classification results can vary. For example:

  Moderator transcript: "मैं आपको यह सूचित करना चाहता हूं"

  Model transcript : "मैं आप को यह सूचित कर ना चाहता हूं"

  The small differences like these are causing changes in the accuracy since the token length and semantic meaning is changed.

7. Deployment Strategy

The service can be deployed as a Rest API on the CRM server.

7.1 ASR service Pricing

ASR Service	API	Pricing
Sarvam AI	Speech to text	Rs 30/hr
Assembly AI	Speech to Text	$0.12/hr

7.2 LLM Inference API Pricing

Groq API Pricing(LINK)

For the best model, the pricing will be $ 0.59 per 1 Million tokens as input (input means prompt+transcription+keywords data), LLM Output pricing $ 0.79 per 1 Million token output.

8. Way forward to Improve performance of models

I. Creation of unskewed knowledge-base covering all categories with human involvement.
II. Grading the performance of model transcription and classified result determined by the model and flagging it wrong if incorrect, by a human expert.
III. Adding newly discovered commonly used keywords by end-users of each category into the keywords database.
IV. Maintaining keywords in a structured format. For example: while mentioning in-combination words it is better to separate words like this **नंबर दीजिए, नंबर चाहिए** instead of mentioning it like this **नंबर-दीजिए/चाहिए**
V. Bi-weekly or monthly update of keywords knowledge base.
VI. Tuning the prompts, most importantly definitions/criterias to classify a complaint into a particular category.
VII. Building a human feedback system to create a proper structured database, if we want to train the LLM.
VIII. Build a constitutional AI with a human loop, where there will be a base model and a tuned base model. Tuned Base model should act as the classifier model and the Base Model as the master for every correct classification the tuned-base-model should be rewarded and retrained on the newly generated data.
IX. Partner with ASR service to share possible data for improving the quality of its response if the call data can be shared. Keeping in mind that if we share data then we don’t share personal information of the end user with call data.